

AI Usage Note

Project Title

IncidentIQ – AI Powered Incident Intelligence & Root Cause Analysis Generator

AI Tools Used

The following AI tools were utilized during the development of this project:

- Gemini API
- Groq API (Fallback AI Provider)
- ChatGPT
- Antigravity AI Development Platform

Purpose of AI Usage

Artificial Intelligence was used as a development assistant to accelerate software development, improve productivity, and support intelligent RCA generation.

1. Frontend Development

AI-assisted development was used to generate:

- React components
- User interfaces
- Responsive layouts
- Dashboard screens
- Report visualization pages
- Chatbot interface
- Role-based navigation

2. Backend Development

AI-assisted development supported:

- Express.js API creation
- SQLite database integration
- JWT authentication
- Role-based access control
- RCA report management APIs
- Manager approval workflows

3. RCA Report Generation

AI was used to analyze:

- Incident Timelines
- Error Logs
- Git Diff File

Based on these inputs, the system generates:

- Incident Summary
- Timeline Analysis
- Error Analysis
- Git Diff Analysis
- Root Cause
- Impact Assessment
- Severity Classification
- Confidence Score
- Recommended Fixes
- Prevention Actions

4. Documentation Support

AI assisted in creating:

- Technical Documentation
- Architecture Descriptions
- Sample Datasets
- Test Cases
- Project Reports

Challenges Encountered During AI-Assisted Development

During development, the following challenges were identified and resolved:

- UI consistency issues required manual refinement.
- Authentication workflows required additional validation.
- AI-generated confidence scores initially appeared unrealistic.
- Gemini API quota limitations required implementation of a fallback AI provider.
- Certain generated UI components required manual optimization for usability and performance.

Best Prompts Used

Prompt 1

Analyze incident timeline, logs and git diff and generate a structured RCA report with root cause, impact assessment, severity, recommended fixes and prevention actions.

Prompt 2

Build a role-based Incident Management system using React, Express.js and SQLite.

Prompt 3

Design a modern SaaS dashboard with enterprise-grade UI and premium user experience.

Prompt 4

Implement multi-provider AI architecture using Gemini as primary provider and Groq as fallback provider.

Human Contribution

The development team was responsible for:

- Requirement Analysis
- System Design
- Prompt Engineering
- UI/UX Validation
- Feature Selection
- Testing
- Debugging
- Integration
- Deployment
- Documentation

AI Usage Declaration

AI tools were used as development assistants for code generation, UI prototyping, RCA report generation, and documentation support. All AI-generated outputs were reviewed, validated, modified, tested, and integrated manually by the development team.